Question 3 Convolutional Architectures (12 points)

Consider the convolutional neural network defined by the layers in the left column below. Fill in the shape of the output volume and the number of parameters at each layer. You can write the shapes in the numpy format (e.g. (128,128,3)).

Notation:

- CONV5-N denotes a convolutional layer with N filters with height and width equal to 5. Padding is 2, and stride is 1.
- POOL2 denotes a 2x2 max-pooling layer with stride of 2 and 0 padding.
- FC-N denotes a fully-connected layer with N neurons

Layer	Activation Volume Dimensions	Number of parameters
Input	$32 \times 32 \times 1$	0
CONV5-10		
POOL2		
CONV5-10		
POOL2		
FC10		

Solution:

Layer	Output Volume Shape	Number of parameters
Input	$32 \times 32 \times 1$	0
CONV5-10	$32 \times 32 \times 10$	$10 \times (5 \times 5 \times 1 + 1)$
POOL2	$16 \times 16 \times 10$	0
CONV5-10	$16 \times 16 \times 10$	$10 \times (5 \times 5 \times 10 + 1)$
POOL2	$8 \times 8 \times 10$	0
FC10	10×1	$10 \times (8 \times 8 \times 10 + 1)$